Cloud Retail Data Lake & Data Warehouse

Problem Statement

A retail organization has multiple operational datasets but needs a scalable cloud architecture for storing raw data, processing it, and providing structured analytical data to business users.

Project Objective

Build a small-scale Azure data platform using a **data lake** → **processing** → **warehouse** architecture.
Azure's architecture guidance describes Data Lake Storage and Synapse as components that can work together for storing, processing, and querying data at scale. *(Microsoft Learn)*

Key Features / Scope

- Upload raw CSV/JSON/Parquet data to Azure Data Lake Storage Gen2
- Organize data into raw/processed/curated zones
- Query lake files using Synapse serverless SQL
- Transform data using Synapse Spark/PySpark
- Convert appropriate datasets to Parquet
- Create a dimensional model:
  - Fact Sales
  - Dim Product
  - Dim Customer
  - Dim Store
  - Dim Date
- Create Synapse external tables/views
- Load curated data into a warehouse
- Implement basic data-quality checks
- Apply appropriate access controls
- Document the architecture
- Monitor pipeline execution

Expected Deliverables

- Azure architecture diagram
- ADLS folder structure
- PySpark notebooks
- Synapse SQL scripts
- Dimensional model
- Data-quality validation
- Sample analytical queries
- Pipeline screenshots
- Technical README

Optional Extension

Add **Delta Lake** for transactional/ACID processing and optimize selected datasets using partitioning/Z-ordering.

Project Plan — Cloud Retail Data Lake & Data Warehouse

> **Status:** v1 · **Owner:** Maged Gamal Ahmed (Project Lead) · **Last updated:** 2026-10-16

1. Overview

Objective. Build a small-scale Azure data platform for a retail organisation using a **data lake → processing → warehouse** architecture. Raw files land in Azure Data Lake Storage Gen2 (ADLS), are cleaned and transformed with Synapse Spark (PySpark), and are served to business users through a dimensional model (Fact Sales + Dim Product, Dim Customer, Dim Store, Dim Date).

In scope (from the project proposal)

- Upload raw CSV / JSON / Parquet data to ADLS Gen2
- Organise the lake into `raw` / `processed` / `curated` zones
- Query lake files with Synapse serverless SQL
- Transform data with Synapse Spark / PySpark and convert suitable datasets to Parquet
- Build the dimensional model, Synapse external tables/views, and load a warehouse
- Implement basic data-quality checks, access controls, and pipeline monitoring
- Produce the documentation and deliverables listed in section 5

Stretch goal (optional): Delta Lake (ACID) plus partitioning / Z-ordering on selected datasets. Starts only after feature freeze (<M5:Alaa' Ahmed Kamal>).

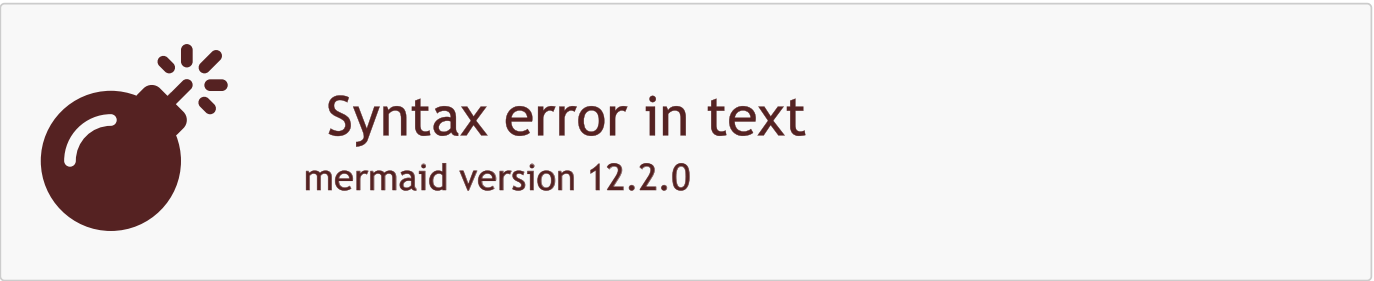
Planning assumptions

Assumption	Value
Team size	6 members
Effort per member	~7 h/week
Duration	~7 weeks (Mon 12 Oct → Sun 30 Nov 2026)
Total capacity	~336 person-hours
Planned task effort	~240 h (+16 h optional); the rest is buffer for meetings, reviews, learning curve and rework
Azure student subscription	One shared subscription with limited credits

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2. Timeline (Gantt)

> Rendered natively by GitHub. In other viewers, paste the block into https://mermaid.live.



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3. Phases

#	Phase	Dates	Goal	Exit criteria
1	Setup & Design	12 – 18 Oct	Everyone can work in Azure and Git; architecture agreed	Architecture v1 approved; all members can open the Synapse workspace and ADLS; repo and board live
2	Ingestion & Data Lake	19 Oct – 1 Nov	Raw data in the lake and queryable	Datasets in `raw/`; serverless SQL queries return correct counts; profiling notes written

#	Phase	Dates	Goal	Exit criteria
3	Transformation	26 Oct – 15 Nov	Clean, typed, Parquet-based `processed` and `curated` zones	PySpark notebooks run end-to-end from a clean state
4	Modeling & Warehouse	9 – 29 Nov	Star schema built and loaded; analysts can query it	Fact Sales + 4 dims loaded; external tables/views work; ≥ 8 sample queries run
5	Quality, Security, Monitoring	16 – 29 Nov	Platform is trustworthy and observable	DQ checks automated; least-privilege access in place; pipeline runs monitored
6	Documentation & Delivery	23 Nov – 02 Dec	Everything reproducible and presentable	README tested from a fresh clone; demo rehearsed

4. Milestones

ID	Date	Milestone	Acceptance criteria	Owner
<M1:Maged Gamal Ahmed>	18 Oct	Design approved	Architecture diagram v1, zone layout, source schema / data dictionary agreed; Azure access verified for all	<M1:Maged Gamal Ahmed>
<M2:Asmaa Mohamed Omar>	28 Oct	Raw zone loaded	All source datasets in ADLS `raw/` (CSV, JSON, Parquet); upload steps documented	<M3:Mohamed Elsaid Abdelrahman Moafi>
<M3:Mohamed Elsaid Abdelrahman Moafi>	15 Nov	Curated zone ready	Cleaned Parquet datasets in `processed/` and `curated/`; notebooks reproducible	<M4: Abdelrahman Adel Mohamed>
<M4:Abdelrahman Adel Mohamed>	25 Nov	Warehouse loaded	Fact Sales and 4 dimensions loaded; external tables/views created; row counts reconcile	<M5:Alaa' Ahmed Kamal>
<M5:Alaa' Ahmed Kamal>	29 Nov	Feature freeze	DQ checks, access control, monitoring and KPI scorecard in place; only docs and bug fixes afterwards	<M1:Maged Gamal Ahmed>
<M6:Esraa Mohamed Khalef>	02 Dec	Final submission	All deliverables complete; demo rehearsed (replace with the real deadline)	<M1:Maged Gamal Ahmed>

5. Deliverables

#	Deliverable	Accountable	Due	Status
1	Azure architecture diagram	<M3:Mohamed Elsaid Abdelrahman Moafi>	v1: <M1:Maged Gamal Ahmed> · final: <M6:Esraa Mohamed Khalef>	
2	ADLS folder structure	<M3:Mohamed Elsaid Abdelrahman Moafi>	<M2:Asmaa Mohamed Omar>	
3	PySpark notebooks	<M4: Abdelrahman Adel Mohamed>	<M3:Mohamed Elsaid Abdelrahman Moafi>	
4	Synapse SQL scripts	<M5:Alaa' Ahmed Kamal>	<M4: Abdelrahman Adel Mohamed>	
5	Dimensional model (ERD + grain + keys)	<M5:Alaa' Ahmed Kamal>	Design: 13 Nov · final: <M4: Abdelrahman Adel Mohamed>	
6	Data-quality validation	<M6:Esraa Mohamed Khalef>	<M5:Alaa' Ahmed Kamal>	

#	Deliverable	Accountable	Due	Status
7	Sample analytical queries	<M6:Esraa Mohamed Khalef>	<M5:Alaa' Ahmed Kamal>	
8	Pipeline screenshots	<M1:Maged Gamal Ahmed> (collected from <M2:Asmaa Mohamed Omar>)	<M6:Esraa Mohamed Khalef>	
9	Technical README	<M1:Maged Gamal Ahmed>	<M6:Esraa Mohamed Khalef>	
10	*(Optional)* Delta Lake extension	<M4: Abdelrahman Adel Mohamed>	After <M5:Alaa' Ahmed Kamal>	

Status key: not started · in progress · done · blocked

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6. Resource Allocation

6.1 People

Member	Role	Owned task hours	Capacity (8 wks)	Utilisation
<M1:Maged Gamal Ahmed>	Project Lead & Documentation	34 h	56 h	61%
<M2:Asmaa Mohamed Omar> > <M1:Maged Gamal Ahmed>	Cloud Infrastructure, Security & Pipeline Ops	35 h	56 h	63%
<M3:Mohamed Elsaid Abdelrahman Moafi> > <M1:Maged Gamal Ahmed>	Ingestion & Data Lake Engineer	42 h	56 h	75%
<M4: Abdelrahman Adel Mohamed> > <M1:Maged Gamal Ahmed>	Transformation Engineer (PySpark)	44 h	56 h	79%
<M5:Alaa' Ahmed Kamal> > <M1:Maged Gamal Ahmed>	Warehouse & Dimensional Modeling Engineer	42 h	56 h	75%
<M6:Esraa Mohamed Khalef> > <M1:Maged Gamal Ahmed>	Data Quality & Analytics Engineer	42 h	56 h	75%

> <M4: Abdelrahman Adel Mohamed>'s work (PySpark) is on the critical path: <M3:Mohamed Elsaid Abdelrahman Moafi> and <M1:Maged Gamal Ahmed> should shadow <M4: Abdelrahman Adel Mohamed> during phase 3 so progress doesn't depend on one person. Optional Delta Lake (+16 h) goes to <M4: Abdelrahman Adel Mohamed> only if the schedule is green after <M5:Alaa' Ahmed Kamal>.

6.2 Involvement by phase

● lead · ○ support · – not involved

Member	P1 Setup	P2 Lake	P3 Transform	P4 Warehouse	P5 Quality/Sec	P6 Docs
<M1:Maged Gamal Ahmed>	●	○	○	○	○	●
<M2:Asmaa Mohamed Omar>	●	○	○	○	●	○
<M3:Mohamed Elsaid Abdelrahman Moafi>	●	●	●	○	–	●
<M4: Abdelrahman Adel Mohamed>	–	○	●	○	○	–
<M5:Alaa' Ahmed Kamal>	○	○	○	●	○	○

Member	P1 Setup	P2 Lake	P3 Transform	P4 Warehouse	P5 Quality/Sec	P6 Docs
<M6:Esraa Mohamed Khalef>	○	●	○	●	●	○

6.3 Azure resources

Resource	Purpose	Sizing / cost control	Owner
Resource group + budget alerts	Single container for all resources	Alerts at 50 / 80 / 100% of credits	<M2:Asmaa Mohamed Omar>
ADLS Gen2 storage account	Data lake (`raw`, `processed`, `curated`)	Hierarchical namespace on; LRS redundancy	<M3:Mohamed Elsaid Abdelrahman Moafi> / <M2:Asmaa Mohamed Omar>
Synapse workspace	Central workspace: SQL, Spark, pipelines, monitoring	One shared workspace	<M2:Asmaa Mohamed Omar>
Serverless SQL pool	Query lake files, external tables/views	Pay per data scanned; query Parquet, avoid `SELECT *` on CSV	<M5:Alaa' Ahmed Kamal>
Apache Spark pool	PySpark transformations	Smallest node size, minimum nodes, **auto-pause ≈ 15 min**	<M4: Abdelrahman Adel Mohamed> / <M2:Asmaa Mohamed Omar>
Warehouse (dedicated SQL pool or alternative)	Serving layer for the star schema	Lowest DWU; **pause when idle**; fallback in the risk register	<M5:Alaa' Ahmed Kamal>
Azure Monitor / Synapse Monitor hub	Pipeline and query monitoring	Default retention	<M2:Asmaa Mohamed Omar>
Key Vault *(optional)*	Secrets instead of keys in code	Only if needed	<M2:Asmaa Mohamed Omar>

6.4 Tools

GitHub (repo, pull requests, Projects board for tasks) · Synapse Studio notebooks / Jupyter · draw.io or Mermaid for diagrams · Faker or a public retail dataset for source data · a team chat channel for async updates.

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7. Cadence & Communication

Event	When	Length	Purpose
Weekly planning	Monday	30 min	Review the board, confirm the week's tasks, check risks
Mid-week sync	Wednesday	15 min	Blockers and hand-offs
Friday demo & retro	Friday	30 min	Show what works; update KPI scorecard; what to change
Async stand-up	Mon / Wed / Fri in chat	2 min	Done · doing · blocked
PR review SLA	Always	—	Reviewed within 24 h

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8. Scope & Change Control

1. Any new scope is raised as a GitHub issue labelled `scope-change`.
2. <M1:Maged Gamal Ahmed> plus at least two other members approve or reject within 48 h.

3. An approved change must name what it replaces or which milestone it moves.
4. The optional Delta Lake extension is **not** started until <M5:Alaa' Ahmed Kamal> is green.

Task Assignment & Roles

> **Status:** v1 · **Owner:** <M1:MAGED GAMAL AHMED> (Project Lead) · **Last updated:** 2026-10-16

1. Team Roster

Label	Name	GitHub handle	Role	Backup (buddy)
	[:MAGED GAMAL AHMED]	*@Migo9000*	Project Lead & Documentation	<M4:ABDELRAHMAN ADEL MOHAMED>
	[:ASMAA MOHAMED OMAR]	*@handle*	Cloud Infrastructure, Security & Pipeline Ops	<M6:ESRAA MOHAMED KHALEF>
	[:MOHAMED ELSAID ABDELRAHMAN MOAFI]	*@Mohamed-Moafi*	Ingestion & Data Lake Engineer	<M5:ALAA' AHMED KAMAL>
	[:ABDELRAHMAN ADEL MOHAMED]	*@handle*	Transformation Engineer (PySpark)	<M1:MAGED GAMAL AHMED>
	[:ALAA' AHMED KAMAL]	*@handle*	Warehouse & Dimensional Modeling Engineer	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>
	[:ESRAA MOHAMED KHALEF]	*@e106sraamohameed-svg*	Data Quality & Analytics Engineer	<M2:ASMAA MOHAMED OMAR>

Each person has a **backup** who understands their work well enough to continue it if they're unavailable. Backups review the owner's pull requests.

2. Role Descriptions

<M1:MAGED GAMAL AHMED> — Project Lead & Documentation

- **Mission:** keep the project on schedule, scope, and documented.
- **Responsibilities:** run the meeting cadence, maintain the board / Gantt / risk register / KPI scorecard, approve scope changes, unblock teammates, own the final demo.
- **Owns:** Technical README, pipeline screenshots, KPI scorecard, documentation consolidation.

<M2:ASMAA MOHAMED OMAR> — Cloud Infrastructure, Security & Pipeline Ops

- **Mission:** a secure, cost-controlled Azure environment that everyone can use.
- **Responsibilities:** provision resource group, ADLS, Synapse workspace and pools; RBAC / ACLs and least privilege; budget alerts and auto-pause; secrets handling; pipeline orchestration and monitoring.
- **Owns:** access-control setup, pipeline triggers and monitoring, role-assignment documentation.

<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI> — Ingestion & Data Lake Engineer

- **Mission:** get data into the lake in an organised, repeatable way.
- **Responsibilities:** select or generate retail datasets and the data dictionary; ADLS folder structure and naming; raw uploads; Parquet conversion and partitioning strategy; architecture diagram.
- **Owns:** ADLS folder structure, architecture diagram (v1 and final), ingestion documentation.

<M4:ABDELRAHMAN ADEL MOHAMED> — Transformation Engineer (PySpark)

- **Mission:** turn raw files into clean, typed, analysis-ready datasets.
- **Responsibilities:** PySpark notebooks for raw → processed → curated; explicit schemas; handling nulls, duplicates, bad dates; conformed dimensions with surrogate keys; optional Delta Lake extension.
- **Owns:** PySpark notebooks.

<M5:ALAA' AHMED KAMAL> — Warehouse & Dimensional Modeling Engineer

- **Mission:** a correct, documented star schema that answers business questions.
- **Responsibilities:** grain statement, ERD, keys; external tables and views; warehouse DDL and load scripts for Fact Sales, Dim Product, Dim Customer, Dim Store, Dim Date.
- **Owns:** dimensional model, Synapse SQL scripts.

<M6:ESRAA MOHAMED KHALEF> — Data Quality & Analytics Engineer

- **Mission:** make sure the data can be trusted and the platform proves its value.
- **Responsibilities:** serverless SQL exploration; profiling; automated DQ checks (row counts, nulls, duplicates, referential integrity); sample analytical queries; supports KPI measurement.
- **Owns:** data-quality validation, sample analytical queries.

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3. RACI Matrix

R = does the work · **A** = accountable / signs off · **C** = consulted · **I** = informed

Activity	<M1:MAGED GAMAL AHMED>	<M2:ASMAA MOHAMED OMAR>	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M4:ABDELRAHMAN ADEL MOHAMED>	<M5:ALAA' AHMED KAMAL>	<M6:ESRAA MOHAMED KHALEF>
Architecture design & diagram	A	C	R	C	C	I
Azure provisioning, RBAC, cost control	A	R	C	C	I	I
Source data & ingestion to raw zone	A	C	R	I	C	C
Serverless SQL exploration	A	I	C	I	C	R
PySpark cleaning & curated transformations	A	C	C	R	C	C
Parquet conversion & partitioning	A	I	R	C	I	I
Dimensional model design	A	I	C	C	R	C
External tables, views,	A	C	C	C	R	I

Activity	<M1:MAGED GAMAL AHMED>	<M2:ASMAA MOHAMED OMAR>	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M4:ABDELRAHMAN ADEL MOHAMED>	<M5:ALAA' AHMED KAMAL>	<M6:ESRAA MOHAMED KHALEF>
warehouse load						
Data-quality checks	A	I	C	C	C	R
Access control	A	R	I	I	C	I
Orchestration & monitoring	A	R	I	C	C	C
Sample analytical queries	A	I	I	I	C	R
KPI tracking	A / R	C	I	I	I	C
README, screenshots, demo	A / R	C	C	C	C	C

4. Task Backlog

Estimates are in hours. Task IDs are used as GitHub issue titles and branch names (e.g. `feature/T12-clean-sales`).

Phase 1 — Setup & Design (12 – 18 Oct)

ID	Task	Owner	Support	Est.	Depends on	Status
T01	Agree scope, roles, meeting cadence	<M1:MAGED GAMAL AHMED>	All	4	—	
T02	Create resource group, ADLS Gen2 account, Synapse workspace	<M2:ASMAA MOHAMED OMAR>	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	6	—	
T03	Configure RBAC for all members and budget alerts	<M2:ASMAA MOHAMED OMAR>	<M1:MAGED GAMAL AHMED>	3	T02	
T04	Repo structure, branch protection, project board	<M1:MAGED GAMAL AHMED>	<M6:ESRAA MOHAMED KHALEF>	4	—	
T05	Architecture diagram v1 and zone layout	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M2:ASMAA MOHAMED OMAR>, <M5:ALAA' AHMED KAMAL>	6	T01	

Phase 2 — Ingestion & Data Lake (19 Oct – 1 Nov)

ID	Task	Owner	Support	Est.	Depends on	Status
T06	Select / generate retail datasets (sales, products, customers,	<M3:MOHAMED ELSAID	<M5:ALAA' AHMED KAMAL>	10	T05	

ID	Task	Owner	Support	Est.	Depends on	Status
	stores) in CSV / JSON / Parquet + data dictionary	ABDELRAHMAN MOAFI>				
T07	Create `raw` / `processed` / `curated` structure and naming conventions	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M2:ASMAA MOHAMED OMAR>	4	T02	
T08	Upload raw data and document the steps	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M6:ESRAA MOHAMED KHALEF>	8	T06, T07	
T09	Serverless SQL exploration scripts over raw files	<M6:ESRAA MOHAMED KHALEF>	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	8	T08	
T10	Data-profiling notebook (nulls, duplicates, types, ranges)	<M6:ESRAA MOHAMED KHALEF>	<M4:ABDELRAHMAN ADEL MOHAMED>	8	T08	

Phase 3 — Transformation (26 Oct – 15 Nov)

ID	Task	Owner	Support	Est.	Depends on	Status
T11	Spark pool setup and ADLS connectivity	<M2:ASMAA MOHAMED OMAR>	<M4:ABDELRAHMAN ADEL MOHAMED>	4	T02, T03	
T12	PySpark raw → processed cleaning notebooks	<M4:ABDELRAHMAN ADEL MOHAMED>	<M6:ESRAA MOHAMED KHALEF>	24	T10, T11	
T13	Convert datasets to Parquet; define partitioning strategy	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M4:ABDELRAHMAN ADEL MOHAMED>	10	T12	
T14	Processed → curated transformations (conformed dimensions, surrogate keys)	<M4:ABDELRAHMAN ADEL MOHAMED>	<M5:ALAA' AHMED KAMAL>	20	T12	

Phase 4 — Modeling & Warehouse (9 – 29 Nov)

ID	Task	Owner	Support	Est.	Depends on	Status
T15	Dimensional model design (grain, ERD, keys)	<M5:ALAA' AHMED KAMAL>	<M4:ABDELRAHMAN ADEL MOHAMED>	10	T10	
T16	Create Synapse external tables and views	<M5:ALAA' AHMED KAMAL>	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	12	T13, T15	
T17	Warehouse DDL and load scripts for Fact Sales + 4 dimensions	<M5:ALAA' AHMED KAMAL>	<M4:ABDELRAHMAN ADEL MOHAMED>	20	T14, T15	
T18	Sample analytical queries (≥ 8)	<M6:ESRAA MOHAMED KHALEF>	<M5:ALAA' AHMED KAMAL>	10	T17	

Phase 5 — Quality, Security, Monitoring (16 – 29 Nov)

ID	Task	Owner	Support	Est.	Depends on	Status
T19	Data-quality checks (row counts, nulls, uniqueness, referential integrity)	<M6:ESRAA MOHAMED KHALEF>	<M4:ABDELRAHMAN ADEL MOHAMED>, <M5:ALAA' AHMED KAMAL>	16	T14, T17	
T20	Access controls: RBAC / ACLs, least privilege, secrets handling	<M2:ASMAA MOHAMED OMAR>	<M1:MAGED GAMAL AHMED>	8	T03	
T21	Pipeline orchestration (triggers) and monitoring	<M2:ASMAA MOHAMED OMAR>	<M6:ESRAA MOHAMED KHALEF>	14	T14, T17	
T22	KPI scorecard and weekly measurement routine	<M1:MAGED GAMAL AHMED>	<M6:ESRAA MOHAMED KHALEF>	6	T21	

Phase 6 — Documentation & Delivery (23 Nov – 02 Dec)

ID	Task	Owner	Support	Est.	Depends on	Status
T23	Final architecture diagram	<M3:MOHAMED ELSAID ABDELRAHMAN MOAFI>	<M2:ASMAA MOHAMED OMAR>	4	T21	
T24	Technical README and docs consolidation	<M1:MAGED GAMAL AHMED>	All	12	T19 – T21	
T25	Pipeline screenshots, demo script, rehearsal	<M1:MAGED GAMAL AHMED>	All	8	T21	

Optional stretch

ID	Task	Owner	Support	Est.	Depends on	Status
T26	Delta Lake conversion; partitioning / Z-ordering on selected datasets	<M4:ABDELRAHMAN ADEL MOHAMED>	<M5:ALAA' AHMED KAMAL>	16	<M5:ALAA' AHMED KAMAL> green	

Total planned effort: ~239 h (+16 h optional).

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5. Working Agreements

Git workflow

- `main` is protected: changes only through pull requests, **at least 1 approval** (the owner's backup where possible).
- Branch names: `feature/<task-id>-<short-name>`, e.g. `feature/T17-load-fact-sales`.
- Small PRs (one task or less), reviewed within 24 h.
- **Never commit secrets** (account keys, SAS tokens, connection strings). Use environment variables or Key Vault; keep a `.gitignore` for credentials and local data.

Notebooks

- Name: `NN_<zone>_<dataset>.ipynb`, e.g. `02_processed_sales.ipynb`.
- Clear outputs before committing (or use `nbstripout`) to keep diffs readable.
- One owner per notebook; others propose changes through PRs.
- Every notebook states its inputs, outputs, and run order in the first cell.

Suggested repo layout

```
.
├── README.md
├── docs/                  # these four documents + architecture diagram
├── notebooks/             # PySpark / profiling notebooks
├── sql/                   # serverless SQL, external tables, warehouse DDL, sample queries
├── data_quality/          # DQ checks and results
├── sample_data/           # small samples only (large data stays in ADLS)
└── screenshots/
```

Definition of Done (every task)

- ☐ Code / SQL committed through a reviewed PR
- ☐ Runs from a clean state without manual fixes
- ☐ Inputs, outputs and assumptions documented (README section or notebook header)
- ☐ No secrets or large data files in Git
- ☐ Board updated and, if relevant, a screenshot saved

Escalation

1. Blocked > 24 h: post in the team channel and tag the buddy and <M1:MAGED GAMAL AHMED>.
2. Unresolved after 48 h: <M1:MAGED GAMAL AHMED> reassigns or adjusts scope, and logs it in the risk register.
3. Disagreement on design: timebox 20 min in the next sync, then <M1:MAGED GAMAL AHMED> decides and records it in `docs/decisions.md`.

Risk Assessment & Mitigation Plan

> **Status:** Draft v1 · **Owner:** <M1:MAGED GAMAL AHMED> (Project Lead) · **Last updated:** 2026-10-16 Review weekly in the Monday planning meeting. Update scores, status, and add new risks as they appear.

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1. Method

Score = Likelihood × Impact, each rated 1 – 5.

Rating	Likelihood	Impact
1	Rare	Negligible: no schedule effect
2	Unlikely	Minor: < 2 days of delay, workaround is easy
3	Possible	Moderate: a milestone slips a few days
4	Likely	Major: a milestone is missed or a deliverable is degraded
5	Almost certain	Severe: project goal at risk

Score	Level	Response
15 – 25	🔴 High	Mitigation starts now; reviewed every week
8 – 14	🟠 Medium	Mitigation planned and owned; reviewed every 2 weeks
1 – 7	🟢 Low	Monitor

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2. Risk Register

Sorted by score. **L** = likelihood, **I** = impact.

ID	Risk	Category	L	I	Score	Level	Mitigation (prevent / reduce)	Contingency (if it happens)	Owner
R01	Azure costs exceed available credits (Spark or warehouse compute left running)	Cost	4	4	16	🔴	Budget alerts at 50 / 80 / 100%; Spark auto-pause ≈ 15 min; pause the warehouse pool when idle; smallest sizes; end-of-day shutdown checklist	Stop non-essential compute; use serverless SQL + CETAS as the serving layer; delete unused resources	M2
R02	Subscription limits, quota or region restrictions block a service (Spark pool, dedicated SQL pool)	Technical	3	5	15	🔴	Provision every resource in week 1 as a smoke test; verify quotas; choose a region with availability	Switch the warehouse to Azure SQL Database or serverless SQL external tables; run Spark locally against exported files	M2
R08	Credentials (account keys, SAS tokens, connection strings) committed to Git, especially if the repo is public	Security	3	5	15	🔴	`.gitignore`; Azure AD / managed identity instead of keys; short-lived SAS; secret scanning (e.g. gitleaks) as a pre-commit hook; PR checklist	Rotate the exposed secret immediately; remove it from Git history; review access logs	M2
R04	Skill gaps in Synapse, PySpark, or dimensional modeling	People	4	3	12	🟠	Week-1 learning tasks (Microsoft Learn modules); buddy system; pair on first notebook / first table	Reassign tasks to stronger members; reduce scope of non-critical features	M1
R05	A team member becomes unavailable (exams, illness, other commitments)	People	3	4	12	🟠	Backup named for every role; small PRs; documented work; shared board	Backup takes over; drop or defer optional scope	M1
R06	Scope creep (extra datasets, Delta Lake, dashboards)	Scope	4	3	12	🟠	MVP defined in the project plan; change control through `scope-change` issues; Delta only after M5	Freeze scope; move new items to a "later" list	M1
R09	Source data quality problems (nulls, duplicates, schema drift, bad	Data	4	3	12	🟠	Profile data in phase 2; explicit schemas in Spark; quarantine bad records instead	Fix at the transformation layer; regenerate	M6

ID	Risk	Category	L	I	Score	Level	Mitigation (prevent / reduce)	Contingency (if it happens)	Owner
	dates) break the pipeline						of failing the whole run; DQ checks at each zone	data if it was synthetic	
R10	Permission / connectivity failures between Synapse, Spark, and ADLS (missing role assignments)	Technical	4	3	12	🟠	Assign storage roles early (T02, T03, T11); connectivity smoke test; keep a role-assignment table	Escalate to M2 same day; use a temporary fix documented in an issue and revert it	M2
R11	Dimensional modeling mistakes (wrong grain, orphan keys, duplicate dimension rows)	Technical	3	4	12	🟠	Written grain statement; team review of the ERD before loading; surrogate keys; referential-integrity DQ tests	Rebuild affected tables from curated data (the pipeline is re-runnable)	M5
R14	Documentation is incomplete or work isn't reproducible near the deadline	Delivery	3	4	12	🟠	Documentation is part of Definition of Done; fresh-clone test of the README in week 7; docs freeze 2 days before submission	Dedicate the last week to documentation; cut optional work	M1
R15	Demo fails (paused pools, expired access, live query errors)	Delivery	3	4	12	🟠	Dry run 48 h before; resume compute ahead of time; screenshots saved after every milestone	Present from the recorded demo and screenshots	M1
R03	No suitable retail dataset, or licensing issues	Data	3	3	9	🟠	Decide dataset (public or synthetic) by end of week 1; agree on a data dictionary	Generate synthetic data with Faker following the agreed schema	M3
R16	Unclear ownership or poor communication	People	3	3	9	🟠	RACI matrix; weekly cadence; board with one owner per task	M1 re-assigns and clarifies in the next sync	M1
R07	Notebook merge conflicts and noisy diffs	Process	4	2	8	🟠	Branch per task; one owner per notebook; clear outputs before commit; small PRs	Resolve in a pairing session; re-run notebooks from clean	M1
R12	Slow queries or Spark jobs (small-files problem, cold starts)	Performance	3	2	6	🟢	Parquet; partition by date; avoid many tiny files; select only needed columns	Repartition / coalesce; materialise heavy joins	M4

ID	Risk	Category	L	I	Score	Level	Mitigation (prevent / reduce)	Contingency (if it happens)	Owner
R13	Delta Lake or Z-ordering not supported by the available Spark runtime	Technical	3	2	6	🟢	Check the Delta version of the Spark pool early; keep it optional	Use partitioning and file compaction instead; drop Z-ordering	M4

3. Top Risks to Watch

1. **R01 Cost.** The most likely cause of a stalled project. Check spend every Friday and run the shutdown checklist.
2. **R02 Platform limits.** Test every Azure service in week 1. Finding a quota problem in week 6 would be expensive.
3. **R08 Secrets in Git.** Easy to cause, hard to undo. Enforce the PR checklist and a secret scanner from day one.

4. Risk Heat Map

	Impact 1	Impact 2	Impact 3	Impact 4	Impact 5
Likelihood 5					
Likelihood 4		R07	R04, R06, R09, R10	R01	
Likelihood 3		R12, R13	R03, R16	R05, R11, R14, R15	R02, R08
Likelihood 2					
Likelihood 1					

5. Review Process

- **Weekly (Monday):** 10 minutes. Re-score, update owner notes, close mitigated risks, add new ones.
- **Triggers for immediate review:** a missed milestone, spend above 80% of budget, a member unavailable for > 3 days, or a failed pipeline run lasting > 1 day.
- **Where to log:** this file for the register; GitHub issues labelled `risk` for actions that need tracking.

Risk change log

Date	Risk	Change	By
2026-10-10	All	Initial register created	M1

Key Performance Indicators (KPIs)

> **Status:** Draft v1 · **Owner:** <M1:MAGED GAMAL AHMED> (Project Lead), measured with M6 · **Last updated:** 2026-10-16
> Targets are proposed starting points for a small-scale project. Items marked *(baseline)* are confirmed after the first full pipeline run (end of phase 3), then locked.

1. How the Standard KPIs Map to This Project

Common KPI	What it means for a data platform project
Response time	Query response time on serverless SQL and the warehouse; Spark job runtime

Common KPI	What it means for a data platform project
System uptime	Pipeline success rate and availability of the serving layer during agreed working / demo windows (pools are paused outside them on purpose, so 24/7 uptime isn't meaningful here)
User adoption rate	How many business questions the platform answers, how many people can run the sample queries and reproduce the project, and stakeholder feedback

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2. KPI Catalogue

2.1 Delivery

ID	KPI	Definition	Target	Source	Frequency	Owner
DEL-1	Deliverable completion	Deliverables marked done ÷ 9 required	100% by M6	Deliverables table	Weekly	M1
DEL-2	Weekly task completion	Tasks completed ÷ tasks planned for the week	≥ 85%	GitHub Projects board	Weekly	M1
DEL-3	PR review turnaround	Time from PR opened to first review	< 24 h	GitHub	Weekly	M1

2.2 Pipeline Reliability (≈ uptime)

ID	KPI	Definition	Target	Source	Frequency	Owner
REL-1	Pipeline run success rate	Successful scheduled / triggered runs ÷ total runs, last 2 weeks	≥ 95%	Synapse Studio → Monitor → Pipeline runs	Weekly	M2
REL-2	Mean time to recover	Average time from failed run to next successful run	< 4 working hours	Monitor hub + issue log	Per incident	M2
REL-3	Serving-layer availability	Time the warehouse / serverless endpoint is reachable during agreed working windows ÷ window time	≥ 99%	Azure Monitor / manual check log	Weekly	M2
REL-4	End-to-end pipeline runtime	Raw → curated → warehouse on the sample data	≤ 30 min *(baseline)*	Monitor hub	Per release	M2

2.3 Performance (≈ response time)

ID	KPI	Definition	Target	Source	Frequency	Owner
PERF-1	Serverless SQL query time	95th-percentile duration of the sample queries on curated Parquet	≤ 10 s	Monitor → SQL requests	Weekly from M3	M5
PERF-2	Warehouse query time	95th-percentile duration of the sample analytical queries	≤ 5 s	Query duration log	Weekly from M4	M5
PERF-3	Spark job runtime	Runtime per notebook; improvement after optimisation	Baseline, then −20% on the 2 slowest jobs	Monitor → Apache Spark applications	Per release	M4
PERF-4	Storage reduction	Parquet size vs. original CSV for the same dataset	≥ 50% smaller	Storage explorer	Once per dataset	M3

2.4 Data Quality

ID	KPI	Definition	Target	Source	Frequency	Owner
DQ-1	DQ check pass rate	Passed checks ÷ total checks per run	100% of critical checks; ≥ 98% overall	`data_quality/` results	Every run	M6
DQ-2	Row-count reconciliation	Tables where raw → curated → warehouse counts match or differences are explained	100%	DQ report	Every run	M6
DQ-3	Orphan fact rows	Fact Sales rows with no matching dimension key	0	SQL check	Every run	M5
DQ-4	Key null / duplicate rate	Nulls or duplicates in primary / business keys	0%	DQ report	Every run	M6
DQ-5	Data freshness	Time between a raw load and updated warehouse tables	≤ 1 h *(baseline)*	Pipeline timestamps	Per release	M2

2.5 Usability & Adoption (≈ user adoption)

ID	KPI	Definition	Target	Source	Frequency	Owner
USE-1	Business questions answered	Distinct analytical queries that run successfully on the warehouse (sales by product, store, customer, date, ...)	≥ 8	`sql/` folder	Weekly from M4	M6
USE-2	Reproducibility	Team members / reviewers who follow the README on a fresh setup and reproduce the results	≥ 2 reviewers outside the owner	Review log	Once, week 7	M1
USE-3	Documentation coverage	Deliverables with a documented README section or notebook header ÷ total	100%	Checklist	Weekly	M1
USE-4	Stakeholder feedback	Average demo score (clarity, usefulness, completeness)	≥ 4 / 5	Feedback form	After demo	M1

2.6 Security & Cost

ID	KPI	Definition	Target	Source	Frequency	Owner
SEC-1	Secrets in repository	Credentials found in Git history	0	Secret scanner	Every PR	M2
SEC-2	Access-control coverage	Containers, pools, and databases with least-privilege roles assigned ÷ total	100%	Role-assignment table	At M5	M2
COST-1	Azure spend vs. budget	Cumulative spend ÷ planned budget	≤ 100%; alert at 80%	Cost Management	Weekly	M2
COST-2	Idle compute	Spark / warehouse pools running outside working hours	0 occurrences	Monitor + spot check	Weekly	M2

3. Status Thresholds

Status	Meaning	Action
🟩 Green	Target met	None
🟧 Amber	Within 15% of target, or a single miss	Owner explains in the next sync
🟥 Red	Worse than 15% from target, or any **critical** KPI missed	Raise as a risk; M1 decides on corrective action

Critical KPIs: DEL-2, REL-1, DQ-1 (critical checks), DQ-3, SEC-1, COST-1.

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4. Weekly Scorecard (copy per week)

Week: __ · **Date:** __ · **Updated by:** __

ID	KPI	Target	Actual	Status	Note
DEL-2	Deliverable completion	100% by M6			
DEL-3	Weekly task completion	≥ 85%			
REL-1	Pipeline success rate	≥ 95%			
PERF-1	Serverless SQL p95	≤ 10 s			
PERF-2	Warehouse p95	≤ 5 s			
DQ-1	DQ pass rate	100% critical			
DQ-3	Orphan fact rows	0			
USE-1	Business questions answered	≥ 8			
SEC-1	Secrets in repo	0			
COST-1	Spend vs. budget	≤ 100%			

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5. Measurement Plan

- **When:** M1 and M6 fill the scorecard on Friday after the demo and retro.
- **How:** use Synapse Studio's Monitor hub (pipeline runs, SQL requests, Spark applications), Azure Cost Management, the GitHub Projects board, and DQ output saved to `data_quality/`.
- **Baselines:** record REL-4, DQ-5, and PERF-3 after the first end-to-end run; lock targets at M3.
- **Trend, not just status:** keep each week's scorecard in `docs/kpi-history/` so the final report can show how the numbers improved.

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6. Definition of Project Success

The project is successful at M6 if:

1. All 9 required deliverables are complete (DEL-2).
2. The pipeline runs end-to-end reliably (REL-1) and produces trusted data (DQ-1, DQ-2, DQ-3).
3. At least 8 business questions are answered from the warehouse (USE-1) within the performance targets (PERF-1, PERF-2).
4. No secrets leaked and access is least-privilege (SEC-1, SEC-2).
5. Spend stayed within budget (COST-1).
6. A reviewer outside the owner can reproduce the work from the README (USE-2).

Literature Review — Cloud Retail Data Lake & Data Warehouse

> **Status:** v1 · **Owner:** *[MAGED GAMAL AHMED]* · **Last updated:** 2026-10-16 · **Sources accessed:** 2026-10-16

1. Purpose, Scope & Method

Purpose. Place our project (an Azure data lake → processing → warehouse platform for retail data) in the context of existing solutions, learn from their strengths and weaknesses, and show how our design differs.

Our project in one paragraph. Raw CSV / JSON / Parquet retail files are stored in Azure Data Lake Storage Gen2 (ADLS) organised into raw / processed / curated zones, explored with Synapse serverless SQL, transformed with Synapse Spark (PySpark), modelled as a star schema (Fact Sales + Dim Product, Customer, Store, Date), and loaded into a warehouse, with data-quality checks, access control, and pipeline monitoring. Delta Lake with partitioning / Z-ordering is an optional extension.

How sources were found.

Step	What was done
Search venues	Microsoft Learn (Architecture Center, Cloud Adoption Framework, Synapse and Fabric docs), GitHub, VLDB / CIDR proceedings, arXiv and Springer, publisher pages
Example queries	"Azure Synapse end-to-end analytics architecture", "data lake zones and containers", "Delta Lake VLDB 2020", "Lakehouse CIDR 2021", "Azure data engineering project retail ADLS Synapse star schema github", "Synapse runtime Spark 3.5 Delta", "Fabric vs Synapse"
Inclusion criteria	Relevant to Azure lakes / Synapse / dimensional modelling / lakehouse design; publicly accessible; identifiable author or organisation; sufficiently documented to assess
Exclusion	Paywalled content we could not read, undated or anonymous posts, tutorials with no architecture detail

Source tiers (used to weigh evidence)

Tier	Meaning	Examples
1	Official vendor documentation, peer-reviewed papers, standard textbooks	[M1]–[M9], [A1]–[A3], [B1]
2	Open-source project repositories (self-reported, not peer-reviewed)	[P1]–[P3]
3	Analyst or consultancy commentary	[C1], [C2]

2. Background Concepts

Data lakes and zones. A data lake stores large volumes of data in any format; its main risks are weak architecture and weak metadata, which can turn it into an unusable "data swamp" [A3]. Microsoft's cloud-scale analytics guidance recommends organising the lake into layers: a raw layer that is kept immutable, an enriched (standardised) layer, and a curated layer that may hold star schemas or data marts. It also notes that a small deployment can consolidate the layers into one storage account [M3].

Medallion naming. The same idea is commonly named bronze (raw), silver (validated / enriched), and gold (curated). Microsoft's Fabric documentation describes it this way and recommends Delta tables for silver and gold [M8]. We map our zones as **raw = bronze, processed = silver, curated = gold**.

Lakehouse and Delta Lake. The lakehouse argument is that open formats such as Parquet on low-cost storage can support warehouse-style workloads and avoid the staleness, cost, and lock-in of two-tier lake-plus-warehouse systems [A2]. Delta Lake adds ACID transactions, time travel, and faster metadata operations to files in cloud object storage through a transaction log stored as Parquet [A1]. This is the basis of our optional extension.

Dimensional modelling. Kimball and Ross's star-schema approach structures analytical data as fact tables surrounded by descriptive dimensions, with retail sales as one of its standard case studies [B1]. It is the model behind our Fact Sales and four dimensions.

Azure Synapse. Microsoft's reference architecture uses ADLS as the home of data across all stages, Synapse pipelines to land data in the raw layer, serverless SQL and Spark pools for processing, and a SQL pool or the lake for serving [M1].

3. Similar Solutions Reviewed

3.1 Microsoft reference architectures — [M1], [M2], [M4]

What they are. Official Azure Architecture Center scenarios, plus a deployment accelerator on GitHub.

- **[M1] Analytics end-to-end with Azure Synapse** shows ingest → store → process → serve → visualise. ADLS is the central store, pipelines copy relational and semi-structured sources into a raw layer, and both serverless SQL and Spark pools are available for processing. It gives guidance on lake layers, folder structure, and file formats.
- **[M2] Data warehouse scenario** (Architecture Center) takes a different shape: source exports land in a staging area in ADLS, Data Factory loads them incrementally into Synapse staging tables, then transforms them into a common structure. The lake is used mainly as a staging area, and the pipeline authenticates to Synapse with a service principal or managed identity.
- **[M4] Azure Synapse end-to-end accelerator** deploys a reference set of resources: a Synapse workspace, a small three-node Spark pool, a DW100 dedicated SQL pool, a Data Explorer pool, a standard LRS storage account, and a Purview account. Several of these are flagged as cost-incurring.

Strengths	Weaknesses / limitations
Authoritative, maintained, and consistent with Microsoft's best practices	Generic enterprise scenarios, not a worked retail dimensional model
[M1] covers the whole lifecycle and both SQL engines	[M2] treats the lake as staging rather than as an analytical layer
[M4] shows realistic sizing and which resources cost money	[M4] provisions more services (Data Explorer, Purview) than a small student project needs, so cost control matters
Managed-identity authentication is shown ([M2])	Pages describe architecture, not measurable success criteria

Relevance. We follow the [M1] flow and the zone guidance in [M3], but we build the retail model and the quality / KPI layer that the reference material leaves open. We reviewed [M4] from its README only.

3.2 Sales Analytics Dashboard – Azure — [P1]

What it is. A public GitHub project using Synapse, ADLS Gen2, Azure Data Factory, T-SQL / PySpark, and Power BI. It follows the bronze → silver → gold pattern. The gold layer is described as a **snowflake schema with one fact table and six dimension tables**, created with Synapse SQL scripts. Entities are e-commerce style (orders, sellers, reviews, geolocation).

Strengths	Weaknesses / limitations
Clear medallion layering with a plain description of each layer	Snowflake design with six dimensions means more joins for analysts
Complete path from lake to a Power BI dashboard	The README does not list a **date dimension**
Repository includes data, source code, a report file, and the dashboard file	The README does not describe data-quality checks, access control, monitoring, or measurable results
Uses Synapse SQL scripts for modelling, similar to our approach	No description or topics on the repository, and no sign of external use or review

Relevance. Confirms that our stack is a common pattern. It also shows two things to avoid: a missing date dimension and undocumented quality and security.

3.3 DataFusion — [P2]

What it is. An end-to-end pipeline using the AdventureWorksLT sample database on a local SQL Server, Azure Data Factory (with a self-hosted integration runtime), ADLS Gen2, **Azure Databricks** for Spark transformations, **Delta** tables in the silver and gold layers, Synapse **serverless SQL** over gold with views and stored procedures, and Power BI with DirectQuery.

Strengths	Weaknesses / limitations
Most complete treatment of **security and governance**: Azure AD role-based access, least privilege, Key Vault for secrets, encryption	Depends on several separately billed services (Databricks, Data Factory, Synapse, Power BI), which raises cost and complexity for a small team
Uses Delta for silver and gold, giving versioning and ACID behaviour	Source is a small sample database, yet the README describes handling "large datasets" without evidence
Includes validation rules in transformations and Azure Monitor for pipeline tracking	Quality, performance, and testing claims are stated without numbers (no row counts, runtimes, or test results)
Serverless SQL over the lake keeps the serving layer lightweight	The README says data is in "fact and dimension tables" but does not specify the model

Relevance. The closest match to our lake-first approach, and the best model for the security section of our README. We replace Databricks and Data Factory with Synapse Spark and Synapse pipelines to keep everything in one workspace, and we add measurable evidence (KPIs).

3.4 SuperStore Sales Data Engineering — [P3]

What it is. A pipeline that extracts sales data from PostgreSQL, cleans it and builds a **star schema** in Python, uploads the result to **Azure Blob Storage**, loads it into a **Synapse dedicated SQL pool**, and builds Power BI dashboards. Apache Airflow, run in a local WSL environment, orchestrates the process. The dataset has 9,993 transactions from 2019 to 2022.

Strengths	Weaknesses / limitations
Simple, easy to follow, with an ERD and screenshots of the pipeline run and role assignment	Blob Storage is used only as a hand-off, with **no lake zones and no Spark processing**
Star schema with a dedicated warehouse	Small, already-clean dataset (the README says no missing values), so little data-quality work is shown
Orchestration is automated and visible	Orchestration runs from a local environment rather than a managed cloud service
Clear technology list	No monitoring, quality checks, or measurable targets described

Relevance. Shows a minimal warehouse-first pipeline. It highlights what a zoned lake and Spark processing add (handling raw, multi-format, messy data), which is the core of our proposal.

3.5 The lakehouse and Fabric direction — [A1], [A2], [M8], [M9], [M5], [C1], [C2]

What it is. Not a competing project but the direction in which platforms are moving. Lakehouse research argues for open formats and a single platform for analytics [A2], Delta Lake supplies the transactional layer [A1], and Microsoft documents a medallion pattern in Fabric in which silver and gold use Delta tables [M8]. One documented option there keeps bronze and silver in lakehouses and makes gold a **data warehouse**, which is the same shape as our design [M8].

Platform status relevant to us.

- Microsoft documents migration paths from Synapse dedicated SQL pools and Synapse Spark to Fabric [M9].
- The Synapse Spark 3.5 runtime (Spark 3.5, Delta Lake 3.2) enters long-term support on 1 Nov 2027, running to 31 Oct 2028, and Microsoft encourages moving to Fabric Runtime 2.0 [M5].
- The Synapse Spark 3.4 runtime reached end of support on 31 Mar 2026 [M6], so new pools should use 3.5.
- Independent analysts reported in 2025 that Synapse remained supported for now but that customers should not expect to stay on it indefinitely [C1], and that moving workloads to Fabric requires manual changes to notebooks, SQL scripts, and pipelines [C2].

Strengths	Weaknesses / limitations
Open formats and ACID tables reduce lock-in and staleness [A1][A2]	The lakehouse papers come from the Databricks authors and argue a position
Fabric simplifies the number of services to manage [C2]	Synapse (the platform in our proposal) is on a longer-term path toward Fabric [M5][C1]
Official migration guidance exists [M9]	Migration is not automatic for everything [C2]

Relevance. Our proposal specifies Synapse, so we stay on it. We keep the logic portable (PySpark, SQL, Parquet / Delta) and note Fabric as future work.

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4. Comparison With Our Project

Criterion	**Our project**	**Microsoft reference [M1] [M4]**	**Sales Analytics Dashboard [P1]**	**DataFusion [P2]**	**SuperStore [P3]**
Source data	Files: CSV / JSON / Parquet	Relational and semi-structured via pipelines	Multiple sources (unspecified)	SQL Server sample DB	PostgreSQL (CSV-derived)
Lake storage	ADLS Gen2	ADLS Gen2	ADLS Gen2	ADLS Gen2	Blob Storage (hand-off only)
Zones	raw / processed / curated	Layered raw zone and others	Bronze / silver / gold	Bronze / silver / gold	None
Processing	Synapse Spark (PySpark)	Spark pools and pipelines	T-SQL / PySpark in Synapse	Databricks (Spark)	Python ETL + Airflow
Serverless SQL on lake	**Yes**	Yes	Not documented	**Yes**	No
Warehouse / serving	Synapse warehouse, star schema	SQL pool or lake	Gold layer via Synapse SQL	Serverless over Delta gold	Dedicated SQL pool
Model	Star: Fact Sales + 4 dims **incl. Dim Date**	Not applicable	Snowflake: 1 fact + 6 dims, no date dim listed	Fact and dims (unspecified)	Star (ERD shown)
File format	Parquet (+ optional Delta)	Guidance on formats	Not documented	Delta (silver / gold)	Not documented
Orchestration	Synapse pipelines	Synapse pipelines	Data Factory	Data Factory	Airflow (local WSL)
Data quality	Automated checks, tracked as KPIs	Guidance only	"Cleaned, validated" silver layer	Validation rules in notebooks	Cleaning and dedup in ETL
Security	RBAC / ACLs, least privilege, no secrets in Git	Not assessed	Not documented	**RBAC, Key Vault, encryption**	Role assignment shown
Monitoring	Synapse Monitor + KPI scorecard	Not assessed	Not documented	Azure Monitor	DAG run screenshots

Criterion	Our project	Microsoft reference [M1] [M4]	Sales Analytics Dashboard [P1]	DataFusion [P2]	SuperStore [P3]
BI layer	Sample SQL queries	Power BI	Power BI	Power BI	Power BI
Measurable targets (KPIs)	**Yes**	Not applicable	Not documented	Not documented	Not documented
Project management artefacts (plan, RACI, risks)	**Yes**	Not applicable	Not documented	Not documented	Not documented

5. Synthesis

What works across the projects

- Layered storage (bronze / silver / gold) and a star-like model are the common, accepted pattern [M3][M8][P1][P2].
- Keeping everything inside one Azure ecosystem simplifies access control and monitoring [M1][P2].
- Open formats (Parquet / Delta) are the dominant storage choice for processed data [A1][A2][M8][P2].

Recurring weaknesses

1. **Quality, security, and monitoring are thin or undocumented**, apart from [P2].
2. **Cost and complexity** grow when many paid services are stacked [P2][M4].
3. **Modelling gaps**: a missing date dimension and an over-normalised snowflake in [P1]; an unspecified model in [P2].
4. **Weak lake usage**: [P3] and [M2] treat storage as a pass-through rather than an analytical layer.
5. **Platform drift**: Spark runtimes and the Synapse-to-Fabric direction change quickly, and some third-party material is already out of date [M5][M6].

6. How Our Project Differs

1. **Lake-first, multi-format ingestion.** We start from CSV / JSON / Parquet files, not a single relational database as in [P2] and [P3].
2. **Explicit zone design** with an immutable raw zone, following Microsoft's guidance [M3].
3. **Both lake and warehouse access.** Serverless SQL over the lake *and* a warehouse serving layer; [P2] has only the former and [P3] only the latter.
4. **A complete star schema with a date dimension**, designed from a written grain statement, rather than the snowflake in [P1] [B1].
5. **Measured quality.** Automated data-quality checks and a KPI scorecard (see `04-kpis.md`), where the reviewed projects give only statements.
6. **Engineering discipline.** Documented plan, RACI, risk register, and cost controls (see `01`–`03`).
7. **Optional Delta Lake / Z-ordering extension.** Not unique ([P2] uses Delta), but included as a stretch goal [A1].

Honest limits of the difference. The architecture is conventional, so our contribution is mostly in rigour, evidence, and documentation, not novelty. We also deliberately leave out a Power BI dashboard, which all three projects include. The team can add a small dashboard if time allows.

7. Insights That Will Be Applied to Our Design

#	Insight	Source	Decision for our project
1	Synapse Spark 3.4 is out of support; 3.5 is current (Delta Lake 3.2)	[M5][M6] [M7]	Create the Spark pool on **runtime 3.5**; confirm in our pool that OPTIMIZE / Z-ORDER commands run before promising them

#	Insight	Source	Decision for our project
2	Raw data should be immutable; layers can share one account at small scale	[M3]	Keep `raw/` write-once; one storage account with a container per zone
3	The curated layer may hold star schemas	[M3]	`curated/` stores model-ready Parquet for Fact Sales and the four dimensions
4	Bronze / silver / gold is the common naming	[M8]	Document the mapping to raw / processed / curated in the README
5	A documented pattern keeps lake layers and makes gold a warehouse	[M8]	Cite this as the justification for our lake + warehouse design
6	Open formats; Delta adds ACID and time travel	[A1][A2]	Parquet by default; Delta only in the stretch task (T26)
7	Weak metadata turns lakes into swamps	[A3]	Data dictionary (T06), naming conventions (T07), and a README section per dataset; Purview is optional because it adds cost [M4]
8	Dimensional modelling guidance and the retail example	[B1]	Write the grain statement first; include **Dim Date**; keep a star, not a snowflake
9	Least privilege, managed identity, Key Vault	[P2][M2]	RBAC / ACLs (T20); no secrets in Git (R08)
10	Compute services cost money	[M4]	Auto-pause Spark; pause the warehouse when idle (R01)
11	Pipelines can run inside the Synapse workspace	[M1]	Use Synapse pipelines instead of an external orchestrator ([P3]) or a separate Data Factory ([P1][P2])
12	Synapse → Fabric is the platform trajectory	[M5][M9] [C1]	Keep logic portable (PySpark, SQL, Parquet / Delta); list Fabric migration under future work
13	Reviewed projects make claims without numbers	[P2]	Record evidence in the KPI scorecard (row counts, runtimes, pass rates, spend)

8. References

Format: numbered, IEEE-style. All URLs accessed 2026-10-16. Microsoft Learn URLs are shown without a language code, so they open in your own locale.

Official documentation (Tier 1)

- **[M1]** Microsoft, "Analytics end-to-end with Azure Synapse," *Azure Architecture Center*, Microsoft Learn. https://learn.microsoft.com/azure/architecture/example-scenario/dataplate2e/data-platform-end-to-end
- **[M2]** Microsoft, Azure Architecture Center example scenario "data warehouse" (page path `/example-scenario/data/data-warehouse`), Microsoft Learn. https://learn.microsoft.com/en-us/azure/architecture/example-scenario/data/data-warehouse
- **[M3]** Microsoft, "Data lake zones and containers," *Cloud Adoption Framework: cloud-scale analytics best practices*, Microsoft Learn. https://learn.microsoft.com/azure/cloud-adoption-framework/scenarios/cloud-scale-analytics/best-practices/data-lake-zones
- **[M4]** Microsoft Azure, "Azure Analytics End to End with Azure Synapse – Deployment Accelerator," GitHub repository. https://github.com/Azure/azure-synapse-analytics-end2end
- **[M5]** Microsoft, "Azure Synapse Runtime for Apache Spark 3.5," Microsoft Learn. https://learn.microsoft.com/azure/synapse-analytics/spark/apache-spark-35-runtime
- **[M6]** Microsoft, "Azure Synapse Runtime for Apache Spark 3.4," Microsoft Learn (end-of-support notice). https://learn.microsoft.com/azure/synapse-analytics/spark/apache-spark-34-runtime

- **[M7]** Microsoft, "Synapse runtime for Apache Spark lifecycle and supportability," Microsoft Learn. https://learn.microsoft.com/azure/synapse-analytics/spark/runtime-for-apache-spark-lifecycle-and-supportability
- **[M8]** Microsoft, "Understand medallion lakehouse architecture for Microsoft Fabric with OneLake," Microsoft Learn. https://learn.microsoft.com/fabric/onelake/onelake-medallion-lakehouse-architecture
- **[M9]** Microsoft, Microsoft Fabric migration overview ("Migrate from legacy, on-premises, or PaaS platforms"), Microsoft Learn. https://learn.microsoft.com/fabric/fundamentals/migration

Peer-reviewed papers and books (Tier 1)

- **[A1]** M. Armbrust *et al.*, "Delta Lake: High-Performance ACID Table Storage over Cloud Object Stores," *Proc. VLDB Endowment*, vol. 13, no. 12, pp. 3411–3424, 2020. https://vldb.org/pvldb/vol13/p3411-armbrust.pdf
- **[A2]** M. Armbrust, A. Ghodsi, R. Xin, and M. Zaharia, "Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics," in *Proc. 11th Conf. on Innovative Data Systems Research (CIDR '21)*, Jan. 11–15, 2021. https://www.cidrdb.org/cidr2021/papers/cidr2021_paper17.pdf
- **[A3]** P. Sawadogo and J. Darmont, "On data lake architectures and metadata management," *Journal of Intelligent Information Systems*, vol. 56, no. 1, pp. 97–120, 2021. DOI: https://doi.org/10.1007/s10844-020-00608-7 (preprint: https://arxiv.org/abs/2107.11152)
- **[B1]** R. Kimball and M. Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd ed. Indianapolis, IN, USA: John Wiley & Sons, 2013. ISBN 978-1-118-53080-1.

Open-source project repositories (Tier 2)

- **[P1]** Dayyak, "Sales-Analytics-Dashboard-Azure," GitHub repository. https://github.com/Dayyak/Sales-Analytics-Dashboard-Azure *(a search result listed this repository as "Ecom-Data-Platform"; GitHub redirects to the current name)*
- **[P2]** mehtavandit, "DataFusion," GitHub repository. https://github.com/mehtavandit/DataFusion
- **[P3]** ShawonSimon, "SuperStore-Sales-Data-Engineering," GitHub repository. https://github.com/ShawonSimon/SuperStore-Sales-Data-Engineering

Analyst and consultancy commentary (Tier 3)

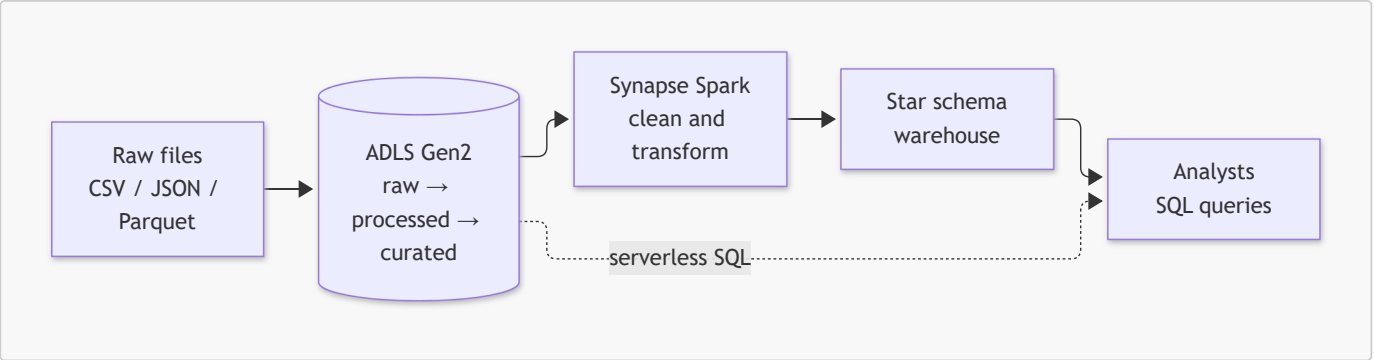
- **[C1]** Directions on Microsoft, "Synapse Analytics Roadmap," Oct. 16, 2025. https://www.directionsonmicrosoft.com/roadmaps/ref/synapse-analytics-roadmap/ *(partly behind a membership wall; we read only the visible summary)*
- **[C2]** B. Smart, "Azure Synapse Analytics vs Microsoft Fabric: side-by-side comparison," *endjin*. https://endjin.com/blog/azure-synapse-analytics-versus-microsoft-fabric-a-side-by-side-comparison *(publication date not confirmed)*

SUMMARY : Requirements Gathering — Cloud Retail Data Lake & Data Warehouse

> **Status:** v1 · **Owner:** M1 (MAGED GAMAL AHMED) · **Last updated:** 2026-10-16

At a Glance

Problem	A retailer has scattered operational data and no scalable way to store raw data, process it, and give business users clean analytical tables.
Solution	A small Azure platform: **data lake** → **processing** → **warehouse**.
Users get	A star schema (Fact Sales + Product, Customer, Store, Date) they can query in seconds.
Must-haves	13 functional requirements, 7 quality requirements
Out of scope	Real-time streaming, machine learning, full BI dashboards (see section 5)



1. Stakeholders & Their Needs

> The retail business users are **simulated** (taken from the problem statement). The team and instructor are real.

Stakeholder	What they care about	What we do about it
Business analysts (main users)	Trustworthy data, simple tables, fast queries	Star schema, data-quality checks, ≥ 8 documented sample queries
Business managers	Answers on sales by product, store, customer, time	Sample queries cover these four views
Future readers (GitHub visitors)	Understand and reuse the work quickly	Architecture diagram, short README, data dictionary

2. Functional Requirements (what the system does)

Priority: **M** = Must · **C** = Could

ID	Requirement	Pri.	Deliverable
FR1	Upload raw CSV / JSON / Parquet to Azure Data Lake Storage Gen2	M	ADLS folder structure
FR2	Organise the lake into **raw / processed / curated** zones	M	ADLS folder structure
FR3	Query lake files with Synapse **serverless SQL**	M	Synapse SQL scripts
FR4	Clean and transform data with **PySpark** (types, nulls, duplicates)	M	PySpark notebooks
FR5	Convert suitable datasets to **Parquet**	M	PySpark notebooks
FR6	Build a **star schema**: Fact Sales, Dim Product, Dim Customer, Dim Store, Dim Date	M	Dimensional model
FR7	Create Synapse external tables and views	M	Synapse SQL scripts
FR8	Load curated data into the warehouse	M	Synapse SQL scripts
FR9	Run basic **data-quality checks** (counts, nulls, duplicates, key integrity)	M	Data-quality validation
FR10	Apply **access controls** (least privilege)	M	Security setup notes
FR11	**Monitor** pipeline runs	M	Pipeline screenshots
FR12	Provide sample analytical queries (≥ 8)	M	Sample queries
FR13	Document the architecture (diagram, README)	M	Diagram, README
FR14	*(Optional)* Delta Lake with partitioning / Z-ordering	C	Extension notebook

3. Quality Requirements (how well it works)

Targets are starting points; they are tracked in `04-kpis.md`.

Area	Requirement	Target	KPI
⚡ **Performance**	Sample queries respond quickly	95% of queries ≤ 10 s (serverless), ≤ 5 s (warehouse); full pipeline ≤ 30 min (baseline)	PERF-1, PERF-2, REL-4
🔁 **Reliability**	Pipeline runs succeed	≥ 95% of runs	REL-1
✅ **Data quality**	Data can be trusted	100% of critical checks pass; 0 orphan fact rows	DQ-1, DQ-3
🔒 **Security**	Only the right people and services have access; no secrets exposed	Least-privilege roles on every container and pool; 0 secrets in Git	SEC-1, SEC-2
😊 **Usability**	Easy to understand and reproduce	A reviewer can follow the README and reproduce results; every deliverable documented; clear table and column names	USE-2, USE-3
💰 **Cost**	Stays within student credits	Spend ≤ budget; compute paused when idle	COST-1
📈 **Scalability / portability**	Design can grow without a rewrite	Parquet, date-based partitioning, logic kept portable (PySpark + SQL)	—

4. Constraints & Assumptions

- **Time and people:** 8 weeks, 6 members, about 7 h/week each.
- **Platform:** Azure Synapse as in the proposal; create the Spark pool on **runtime 3.5**, since 3.4 is out of support.
- **Data:** public or synthetic retail data.
- **Budget:** limited Azure credits, so small pool sizes and auto-pause.

5. Out of Scope (proposed)

Real-time streaming · machine-learning models · a full BI dashboard (a small one can be added if time allows).

6. Definition of Success

The project is accepted when:

1. All **FR1–FR13** are demonstrated.
2. All **quality targets** are green on the final KPI scorecard.
3. A teammate can **reproduce** the results from the README.